

**MATH 1342 – Elementary Statistics – Week 16: Putting It All Together and Final Exam**

**Dec 7 – 9** | Review: Putting It All Together – From Sampling to Hypothesis Testing (Mon Dec 7) • Final Exam – Wed Dec 9, online (Blackboard, proctored) | Fall 2026 · Section 019 · MW 7:30–8:50 PM · RRC Room 8210

| Course-level Objectives                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Module-level Objectives                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Learning Materials                                                               | #             | Activities                                                                                                                                                        | #             | Assessments                                                                    | #             |
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| 3. Find and use empirical probabilities in bootstrap distributions to find confidence intervals and in randomization distributions to test hypotheses.<br>4. Find and use theoretical probabilities from normal, t, chi-squared and F distributions to form confidence intervals and test hypotheses. Apply the 95% rule to normal and to approximately normal distributions.<br>7. Apply ideas of appropriate sampling techniques and experimental design to data production.<br>9. Estimate single means, difference of two means, single proportions and difference of two proportions using confidence intervals. Interpret the results.<br>10. Demonstrate skills in hypothesis testing for means and proportions, for single populations and comparison of two populations.<br>11. Demonstrate skills in hypothesis testing using chi-squared tests.<br>12. Demonstrate skills in inference for regression and ANOVA techniques. | 1. Choose the correct procedure among one- and two-proportion z, one- and two-sample t, paired t, chi-square, ANOVA, and regression, and carry out a complete six-step hypothesis test. (CO9, CO10, CO11, CO12)<br>2. Carry out and interpret a complete confidence interval or hypothesis test, using simulation or theoretical distributions as appropriate. (CO3, CO4, CO9, CO10)<br>3. Interpret results in context, including what the study design allows the conclusion to claim about generalization and causation. (CO7) | Review Slides: Putting It All Together (with answer key and no-answers versions) | MO1, MO2, MO3 | Participation: One Complete Test with a Partner – 10 problems, one per pair (narrator/writer roles), 5-minute presentations; extension: build the matching 95% CI | MO1, MO2, MO3 | Final Exam – Wed Dec 9 (Blackboard, proctored, 2 hours, cumulative; no retake) | MO1, MO2, MO3 |
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